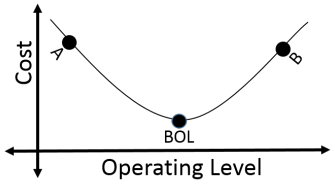
Strategic Capacity Management

Definitions	
Capacity	Limiting capability of a productive unit (system) to produce within a given period of time → units = output/unit time
Peak Capacity	Maximum output that the facility could achieve
Best Operating Level	Capacity for which process was designed. Volume at which avg unit cost is minimized
Effective Capacity	Maximum output that a facility can economically sustain under normal conditions. This is the level of capacity for which the average unit cost is minimized (aka: **Best Operating Level**) - You are not under or over utilizing  BOL: best operating level A: higher overhead cost, higher material cost B: overtime, additional employees
Capacity Cushion	The amount of reserve capacity that a firm maintains to handle sudden increases in demand or temporary losses of production capacity. It measures the amount by which average utilization (in terms of effective capacity) falls below 100%
Capacity Flexibility	Ability to increase or decrease production and/or shift quickly between products requires: flexible plants, flexible processes, flexible workers
Determining Capacity Requirements (steps)	1. Forecast sales within each individual product line 2. Calculate equipment and labor requirement to meet forecast 3. Project equipment and labor availability over planning horizon
Decision Tree Problems	1. Draw the tree with alternatives as branches 2. Expand each branch with the possible states, probabilities, and payoffs 3. Determine the expected value of each alternative decision
Decision Tree Evaluation Criteria	• *Expected Value:* find the expected payoff • *Maxi-min:* maximize the minimum profit (or minimize the maximum loss) • *Maxi-max:* maximize the maximum profit (or minimize the minimum loss) • *Mini-max Regret* (minimize the maximum regret). For each alternate and state, regret = max under that condition among all alternates - value in that condition

Equations	
Capacity Utilization Rate	$Cap\ Util\ Rate = \frac{Average\ Output\ Rate}{Effective\ Capacity} = \frac{Capacity\ Used}{Best\ Operating\ Level}$ $utilization = \frac{hours\ used}{(\#machines) * (theoretical\ hours\ available\ per\ machine)}$
Capacity Cushion	$Cap\ Cushion = 100\% - utilization\ rate$

*understand slides on Capacity Planning Considerations & Frequent/Infrequent Expansion - what do you need to take into account for a holistic view?